

Topic 1.10 - Rational Functions and Holes

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Identify holes from canceled factors and determine exact coordinates.
- Distinguish holes from zeros and vertical asymptotes while preserving the original domain.
- Define repaired piecewise functions and construct rational functions with specified holes.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Find the hole and vertical asymptote of $\frac{(x-2)(x+5)}{(x-2)(x-4)}$.

Question 2

Factoring

No calculator

Find all features of $\frac{x^2-1}{x^2+3x+2}$.

Section A continued

Question 3

Coordinate

No calculator

Determine the hole coordinate of $\frac{(x-3)^2(x+1)}{(x-3)(x+4)}$.

Question 4

Reasoning

No calculator

Explain why $\frac{(x+2)(x-1)}{(x+2)^2}$ has no hole at -2 .

Question 5

Repair

No calculator

Repair $\frac{(x-5)(x+1)}{(x-5)(x+3)}$ at its removable discontinuity.

Section B: Application and reasoning**Question 6**

Construction

No calculator

Construct a function with hole $(0, 2)$ and vertical asymptote 3.

Question 7

Construction

No calculator

Construct a function with hole $(2, 5)$, zero 4, and vertical asymptote -1 .

Question 8

Error analysis

No calculator

A student says the simplified rule and original function are equal at the canceled input. Correct the statement.

Section C: Challenge and error analysis**Question 9**

Conceptual

No calculator

Can a vertical asymptote be repaired by assigning a single output? Explain.

Question 10

Multiple holes

No calculator

Find holes and the vertical asymptote of $\frac{(x-2)(x^2-1)}{(x-2)(x-1)(x+4)}$.

Question 11

Challenge

No calculator

Construct a rational function with two holes, no vertical asymptotes, and simplified rule $x^2 + 1$.

Question 12

Proof challenge

No calculator

Prove that multiplying a rational rule by $(x - a)/(x - a)$ can change the function even though the factor equals 1 wherever defined.

Section D: AP-style multiple choice

MCQ 1

No calculator

The hole of $\frac{(x-2)(x+3)}{(x-2)(x+1)}$ is

- A. (2, 0)
- B. (2, 5/3)
- C. (-1, 0)
- D. vertical asymptote $x = 2$

MCQ 2

No calculator

After a common factor cancels, its zero is

- A. restored to the domain
- B. a removable exclusion
- C. always a zero
- D. always a vertical asymptote

MCQ 3

No calculator

Which value repairs $\frac{x^2-4}{x-2}$ at $x = 2$?

- A. 0
- B. 2
- C. 4
- D. undefined

MCQ 4

No calculator

If cancellation leaves another copy of the same factor in the denominator, the input is generally

- A. a hole
- B. a zero
- C. a vertical asymptote
- D. in the domain

AP-style free response 1**No calculator - 6 points**

Let $R(x) = \frac{(x-1)(x+2)^2}{(x-1)(x+2)(x-3)}$.

A. State exclusions and simplify.

B. Find all hole coordinates.

C. Identify vertical asymptotes and zeros.

D. Define a repaired piecewise function at every hole.

AP-style free response 2**Calculator permitted - 6 points**

Let $M(x) = \frac{(x-2)(x^2-1)}{(x-2)(x-1)(x+4)}$.

A. Use algebra to find all exclusions and simplify.

B. Determine both hole coordinates and the vertical asymptote.

C. Use a graph/table to verify the finite values approached at the holes.

D. Explain why graphing only the simplified expression gives an incomplete representation.
